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Choosing the DC/AC ratio for a hybrid solar system

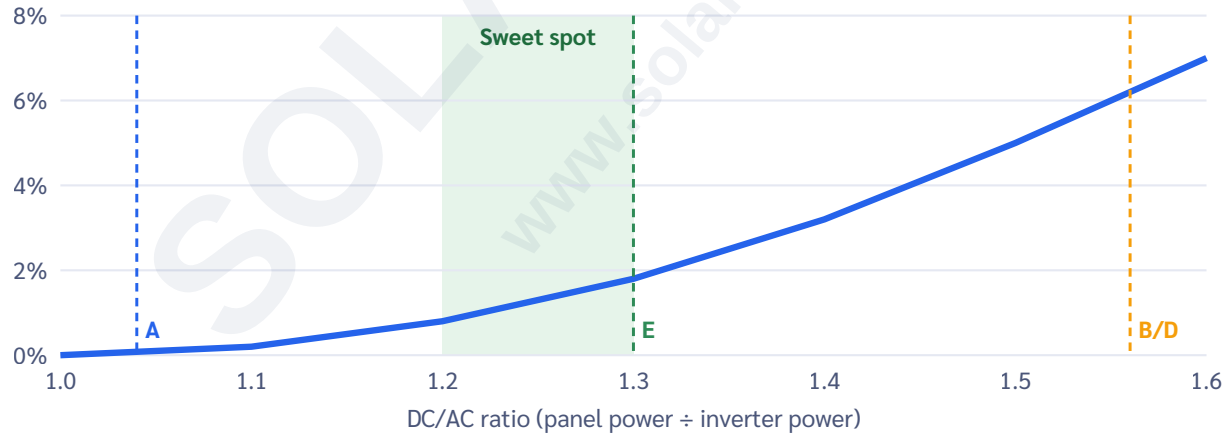
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> **In short:** for a typical grid-tied or hybrid system in Thailand, the best DC/AC ratio is usually about **1.2–1.3**. For a system that cannot export to the grid, though, **how much power the building uses** matters more than the ratio.

Why 1.2–1.3 is the usual sweet spot

- Panels rarely reach their nameplate output in the Thai climate. Heat costs about 0.26% per °C above 25 °C, and cells sit at 55–65 °C on a hot roof, which takes 8–10% off. Dust, cable losses and roof orientation take off a little more.
- So a 10 kW inverter with 10.4 kWp of panels (1.04) seldom runs near full power. The inverter you paid for is under-used most of the day.
- At 1.2–1.3, the inverter runs near full power for more hours, and the power lost by capping the peaks (clipping) is only about 1–2% a year.
- Above about 1.4, clipping grows quickly: roughly 4–7% at 1.5–1.6 in northern Thailand. Each extra panel then earns less.

Approximate energy lost to clipping, northern Thailand



Clipping figures are rough estimates for a south-west roof in northern Thailand, without a battery. Real values depend on weather, tilt and inverter settings.

What's different about a hybrid like the Solis S6

- Solis allows up to 1.6 (16 kW of panels on the 10 kW model, 24 kW on the 15 kW). That's because surplus PV can charge the battery directly when the AC output is at its limit.
- With one 16 kWh battery, charging is capped at about 8 kW. Once the battery is full around midday, the extra panels are clipped again.
- A hybrid can therefore run at 1.3–1.5 sensibly, but only if there is a battery (or load) to take the surplus.

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Example: a zero-export building using about 13,300 kWh a year

Option	DC/AC	Comment
A: 10 kW + 10.40 kWp	1.04	Under-loaded inverter, but the best return because the system matches the building's use
B: 10 kW + 15.60 kWp	1.56	At the Solis limit; extra output mostly unused with zero export
C: 15 kW + 15.60 kWp	1.04	Oversized inverter today; it only pays off if the system is expanded later
D: 15 kW + 23.40 kWp	1.56	At the limit; about 60% of output unused at today's load
E: 10 kW + 13.00 kWp	**1.30**	**The technically ideal size**

When the system cannot export to the grid, a higher ratio mainly produces power nobody uses.

Recommendation

For the building in this example, the technically ideal size is **a 10 kW inverter with about 20 panels (13.0 kWp, ratio 1.30)**, wired as 2 strings of 10 at about 540 V open-circuit on a cold morning.

- It costs more than option A but saves only slightly more on the bill, since the building's load is still the limit.
- It uses the inverter better and leaves headroom if the building's use grows.

For the best return at today's load, option A is still the best value. For a design for your building, book a consultation at www.solaress.org or LINE: solaress1.

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